CPSC 368: Connecting to MongoDB

Connecting to the MongoDB Server

As the department's MongoDB database is a protected host, we must first tunnel into the department before accessing the database. The machine that is trying to connect to the database must keep the tunnel open throughout the duration you want to connect to the database.

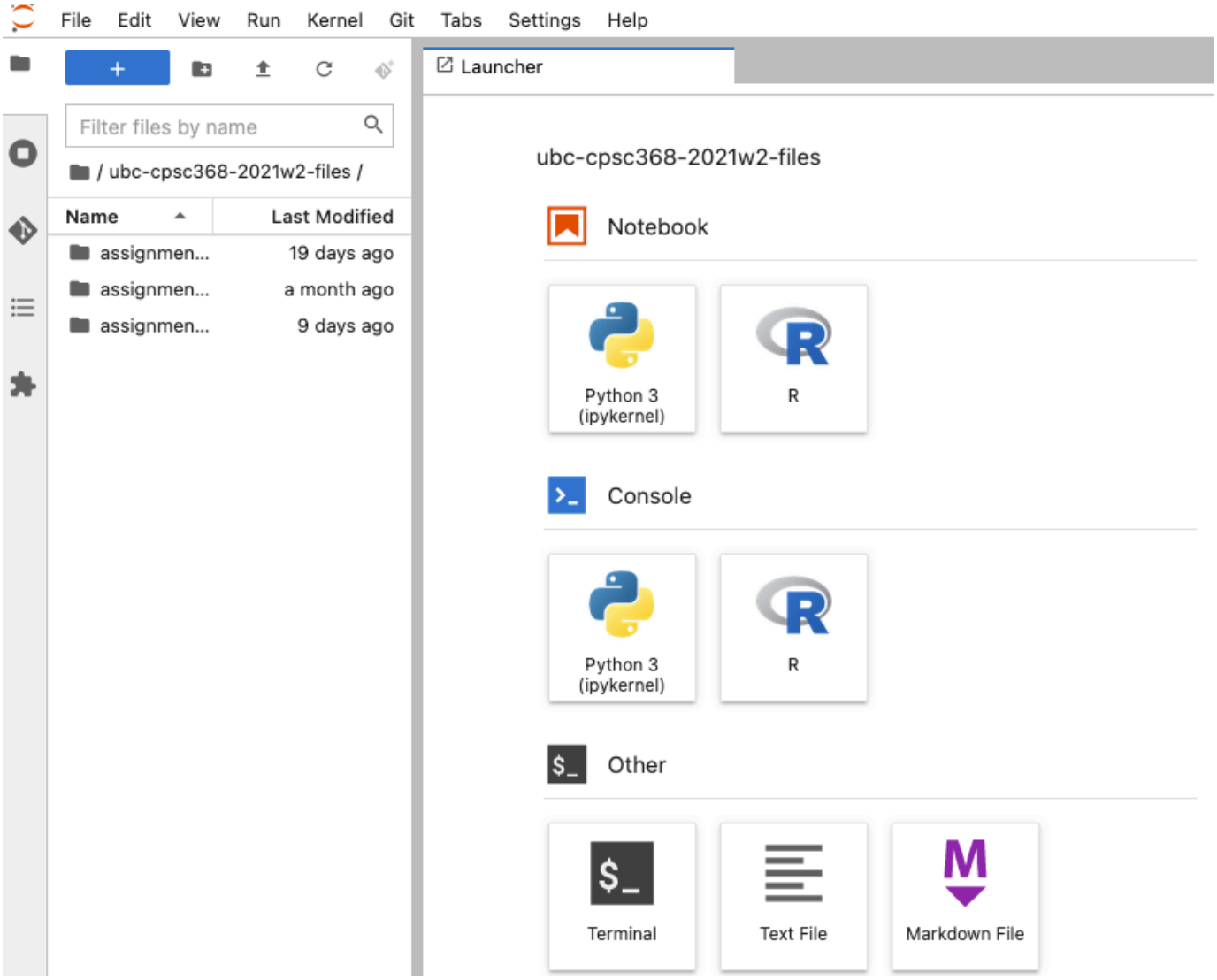
Depending on whether you are working from Syzygy or your own local computer, step 1 will vary. Step 2 will remain the same.

STEP 1: TUNNELLING TO THE DEPARTMENT SERVER

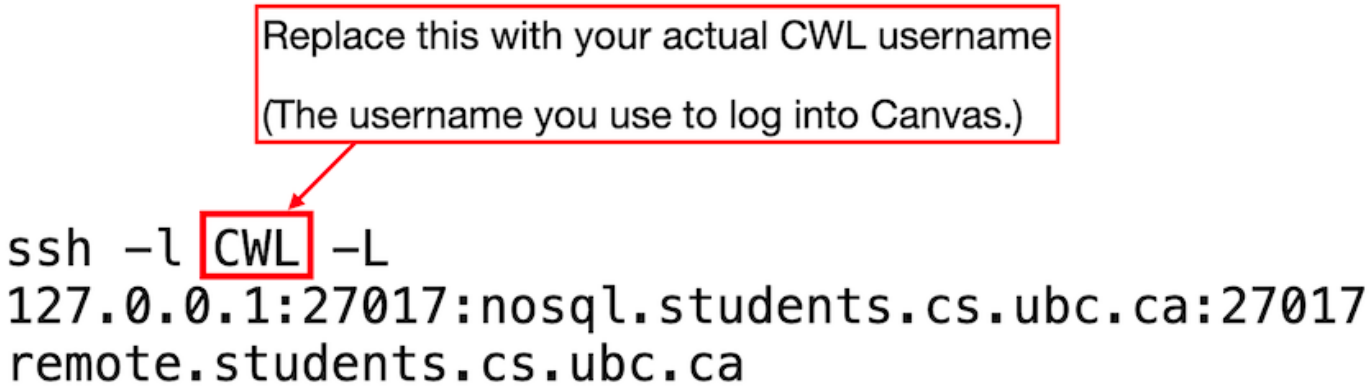
Option 1: If you are working from Syzygy...

1. Go to the launcher view of Jupyter. This is the view with the sidebar that lists all the files in your account (see screenshot below). If you do not see the launcher view of Jupyter,

click on the Jupyter logo in the top left corner of the screen.



2. Click on Terminal.
3. You will need to use a SSH command to connect to the department servers. We have provided the command for you to copy/paste below (see the pink text below the image) but **please replace `CWL` with your actual CWL id.**




```
ssh -l CWL -L 127.0.0.1:27017:nosql.students.cs.ubc.ca:27017 remote.students.cs.ubc.ca
```

4. After clicking enter, you may be prompted with a message that looks like the following:

The authenticity of host 'remote.students.cs.ubc.ca (<some IP address>)' can't be established.
ED25519 key fingerprint is <some series of numbers and letters>.
Are you sure you want to continue connecting (yes/no/[fingerprint])?

Type yes and press enter.

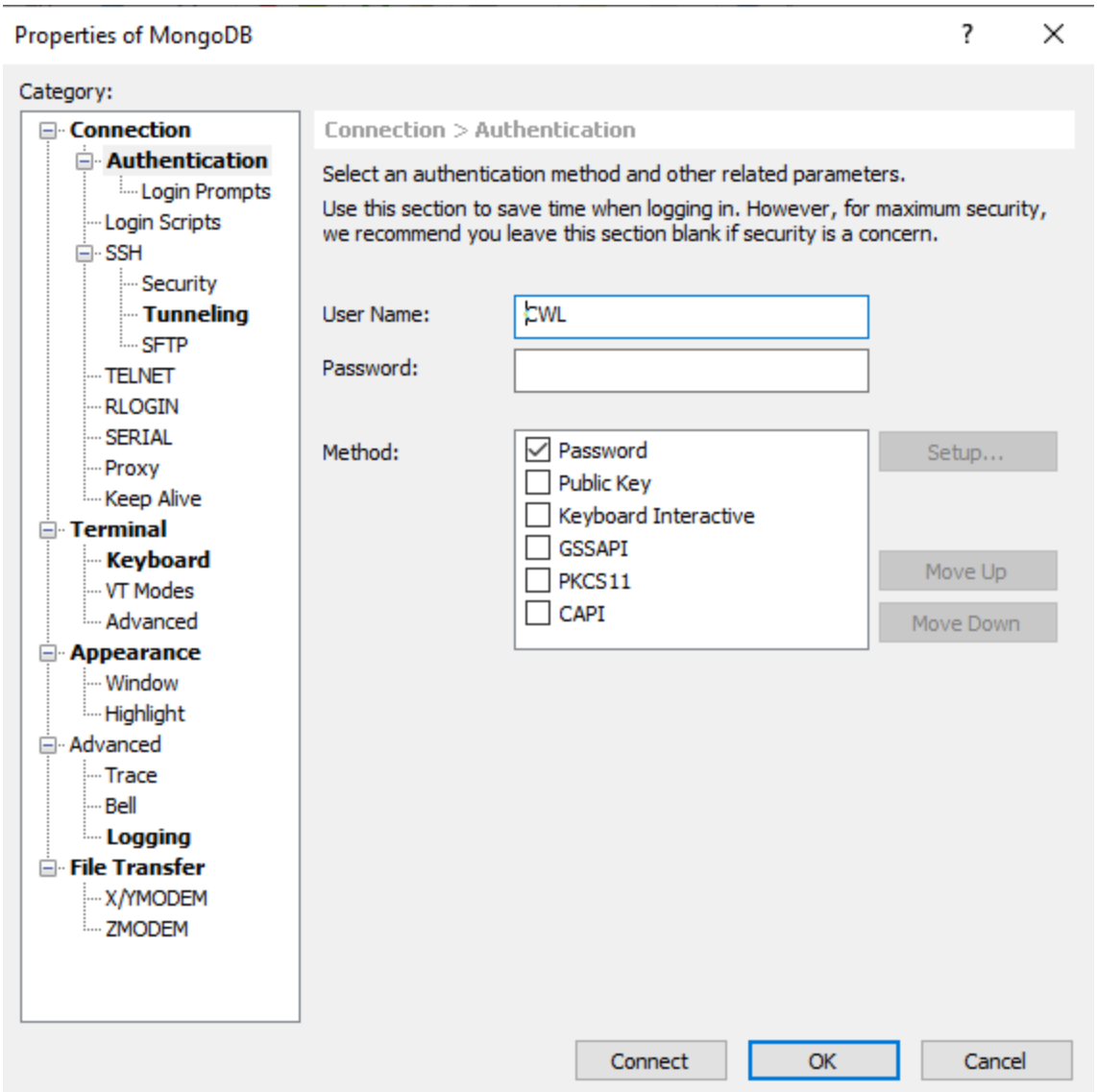
5. You must keep this terminal window open while you access the database. When you are finished or want to stop working, type `exit` into the terminal window and hit enter to close the session.
6. Move to step 2 to test that your connection works.

Option 2: If you are working from a local Jupyter Installation...

This section is for anyone who has installed a version on Jupyter on their computer and wants to use this local installation to access the UBC CS department's MongoDB database. **It assumes that you have already installed Jupyter and Pymongo.** If you have not done these things, please refer to the "Programming Environment" page on Canvas for instructions on how to do so.

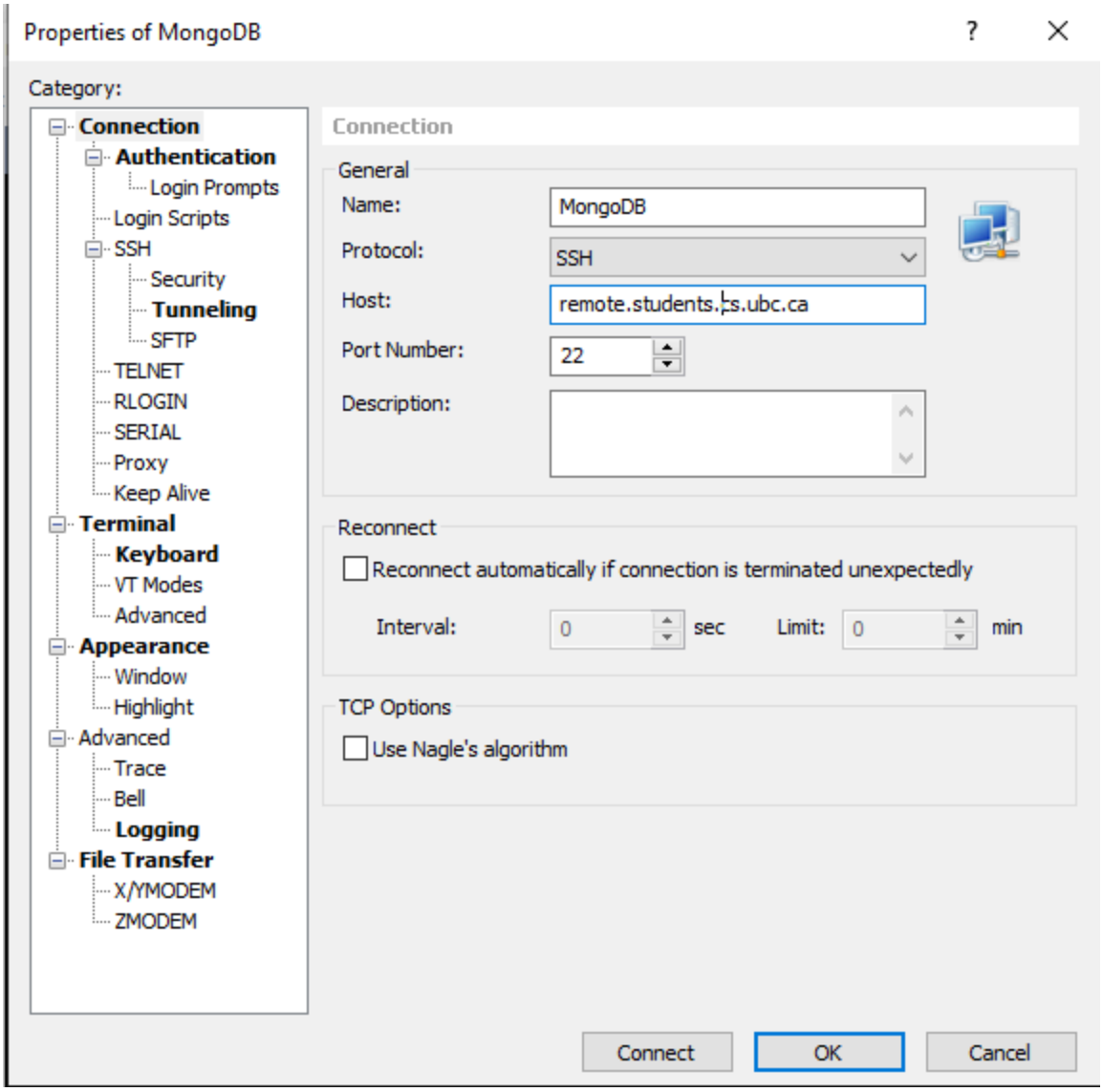
WINDOWS

1. If you do not have XManager installed on your computer, download and install a copy.
2. Create an XShell session in XManager by clicking on the leftmost button (the one with a + sign).
3. In **Connection** set the values as per the screenshot below. Make sure you use **your own CWL username and password**.

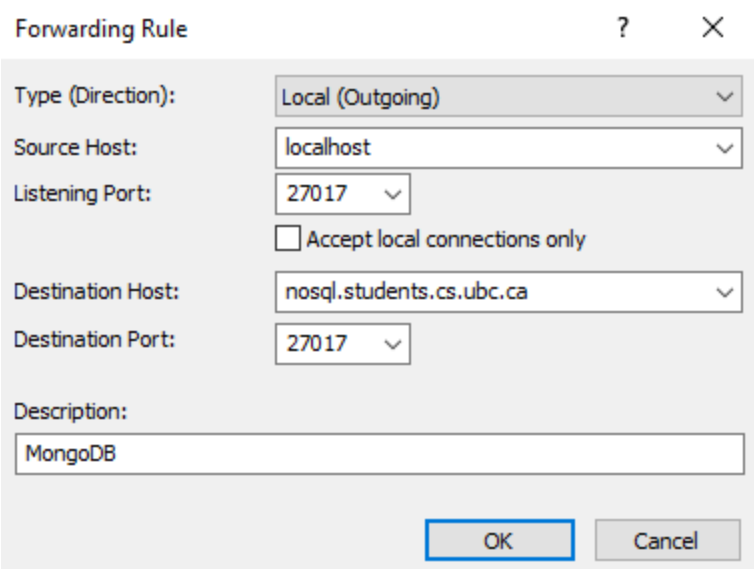


4. In **Authentication**, set the values as per the screenshot below (you can choose to use SSH authentication instead of password but in terms of getting things set up and running

quickly, password might be a better choice).



5. In **Tunneling**, click add and fill in the information as per the screenshot below. When you are finished, click "Ok".



6. Check the "Forward X11 connection" box. Forward the connection to Xmanager. Click "Ok".
7. Launch your connection by clicking on "MongoDB" in the sidebar in XManager.

8. Move to step 2 to test your connection.

MAC OR UNIX

1. Open Terminal and run:
   `ssh -l CWL -L localhost:27017:nosql.students.cs.ubc.ca:27017 remote.students.cs.ubc.ca`

   Be sure to replace CWL with your actual CWL username.

2. Move to step 2 to test your connection.

STEP 2: TEST YOUR CONNECTION

Try running the following in a Jupyter cell. Be sure update the values of the `CWL` and `SNUM` variables with your actual CWL username and student number (you don't need to put an a in front of the student number, just the actual number is fine).

```
import pymongo

CWL = 'Put your CWL here'
SNUM = 'Put your snum here'

if CWL.strip() == "" or CWL == 'Put your CWL here' or SNUM.strip() == "" or SNUM == 'Put your
    print("You need up to update the value of the CWL and/or SNUM variables before proceeding
else:
    connection_string = f"mongodb://{CWL}:a{SNUM}@localhost:27017/{CWL}"
    client = pymongo.MongoClient(connection_string)
    db = client[CWL]["tutorial"]["ubcfood"]

    db.insert_one(
        {
            "name": "Pacific Poke",
        "address": {
                "building_number": 2366,
                "road name": "Main Mall",
            },
            "phone_number": "604.822.1992",
            "labels": ["quick service"],
        "closed": ["Saturday", "Sunday"],
        "meal_plans": ["Chomp Carryover", "Chomp Departments", "Chomp Student", "Chomp UBCcar
        }
    )
```

If all has gone well, you should see something like `pymongo.results.InsertOneResult at 0x10b2cc840` appear underneath the cell that contains the code above.

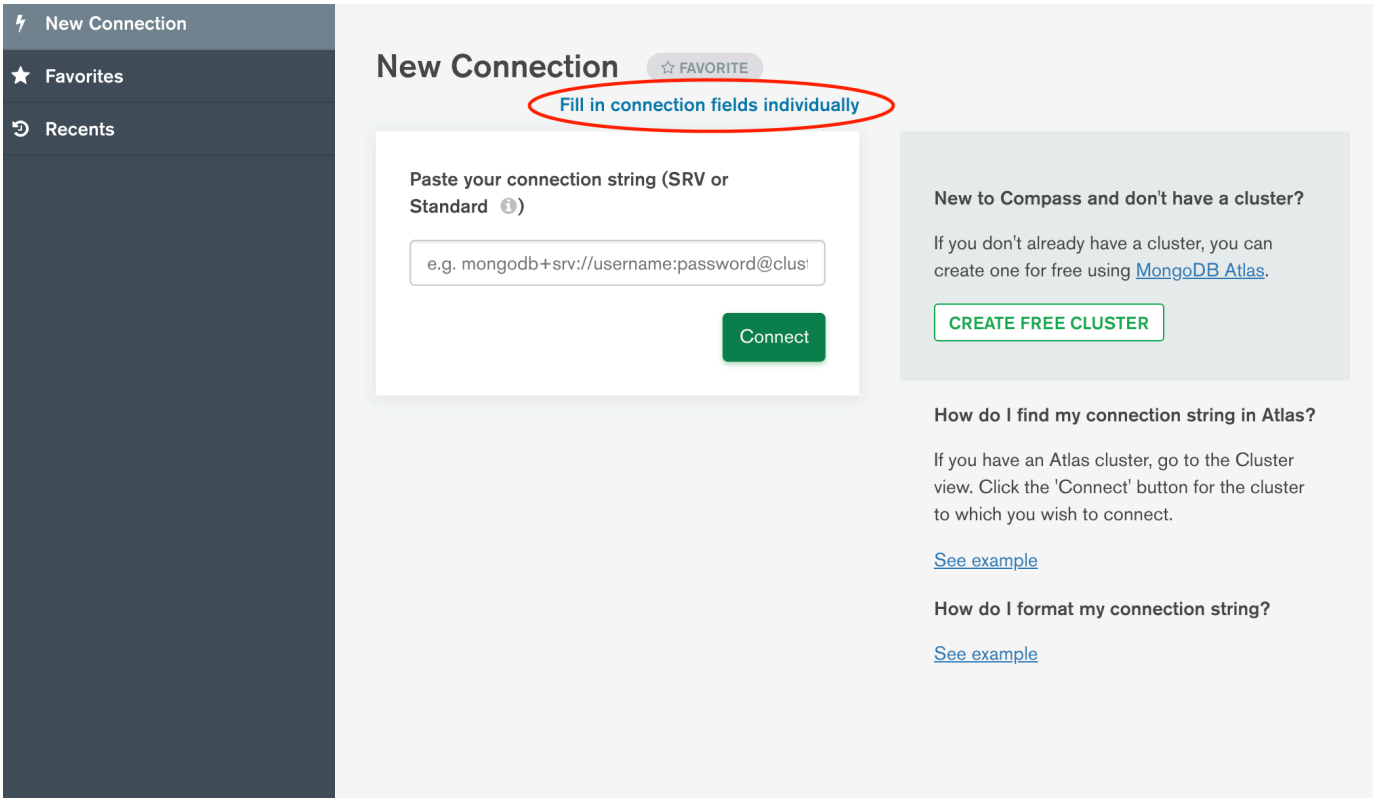
You can also check to see if the insertion went as expected by launching MongoDB Compass.

MONGODB COMPASS

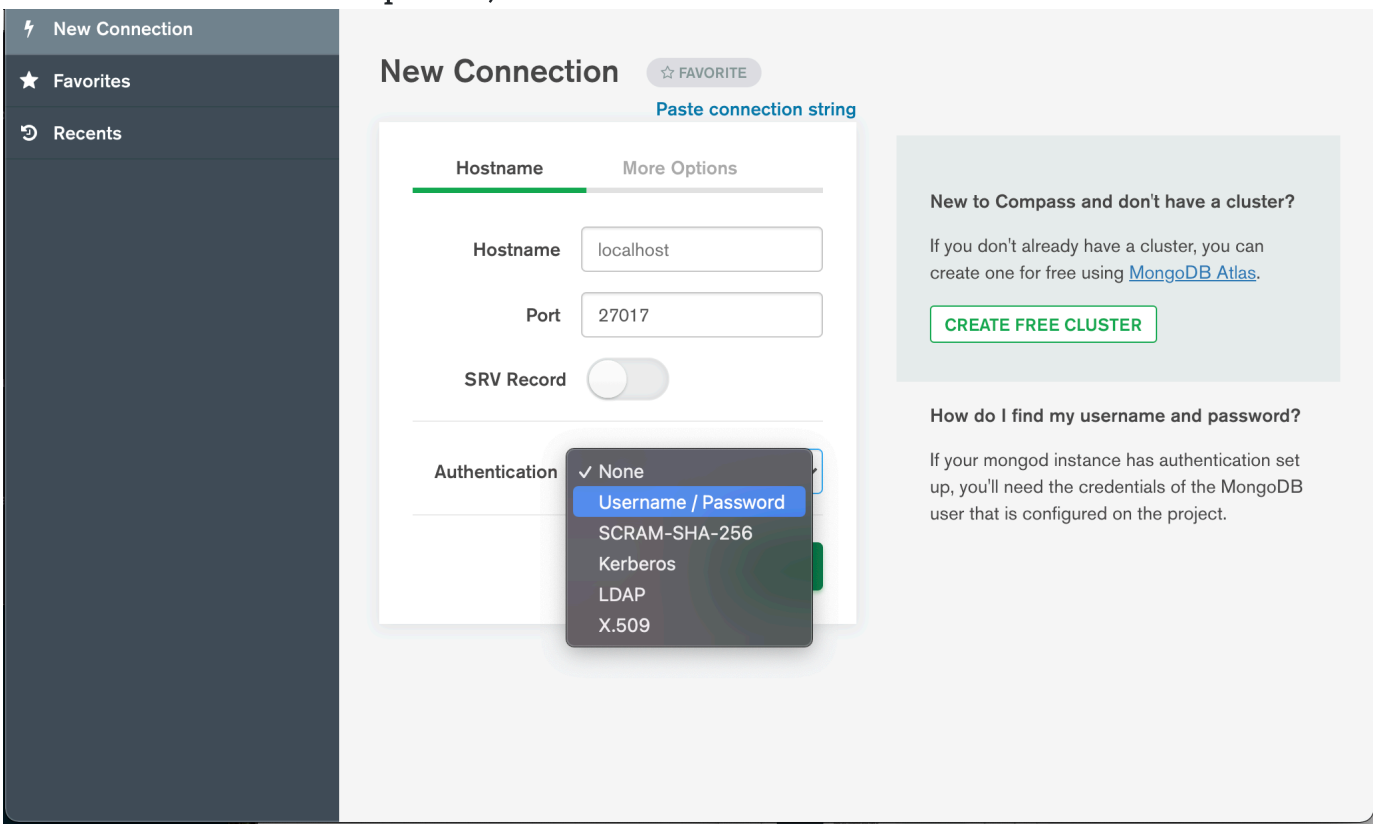
Compass is a graphical user interface that allows you to interact with the database in all the ways your code would allow. We can use it to check that our code has performed the expected action in our database.

The rest of this section proceeds on the assumption that the machine you are using Compass on has already tunneled into the department's server.

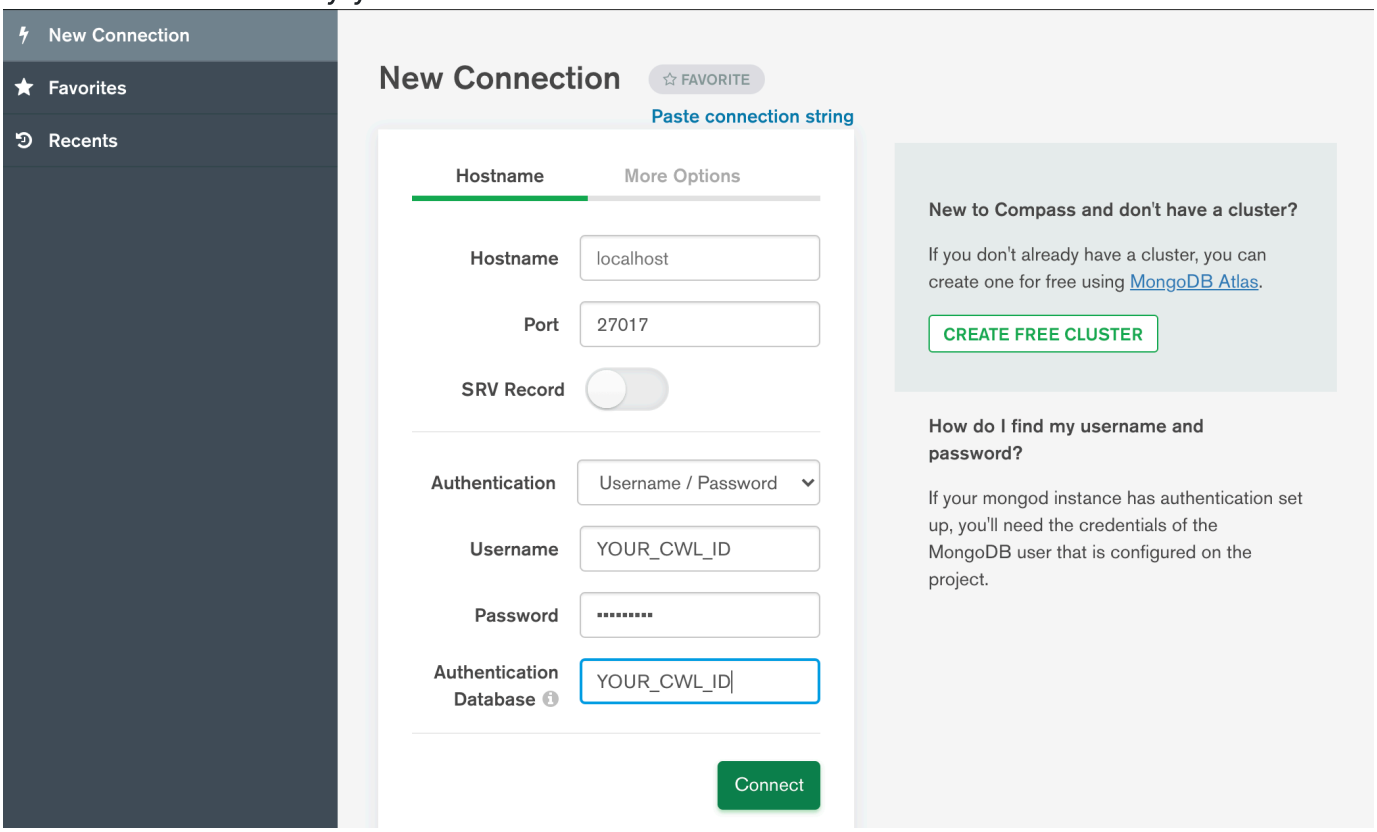
1. First, download and install Compass.
2. Ensure that you have tunneled into the server (either by using the Windows or Unix instructions). You need to tunnel into the department **on the machine that you are running Compass on**. That is, if you are using Syzygy and you want to use MongoDB Compass, you have to tunnel into the department in **two** places (once via Terminal on Syzygy and once on your own computer).
3. Launch Compass and click on the "Fill in connection fields individually" option.



4. In the Authentication dropdown, choose "Username/Password".



5. Fill in the rest of the details according to the screenshot below. The password should be the letter `a` followed by your student number.



6. Click Connect.
7. Once in the application, you should see a collection in your database called `ubcfood`. If you expand that collection, you should be able to see any documents that exist in the database.

8. If you are coming from Step 2, then you should see a document in the `ubcfood` collection that contains information about Pacific Poke.